Empirical Applied Microeconomics Final Project
Post-COVID Labor-Market Outcomes of Mothers in Teleworkable Occupations

김범준

School of Economics, Yonsei University

Outline

Motivation

- Goldin (2014) argues that a large part of the gender wage gap is related to the high cost of temporal flexibility.
- In many high-wage occupations, workers face strong rewards for long and inflexible working hours.
- Since childcare responsibilities are disproportionately borne by mothers, these rigid work arrangements may contribute to the motherhood penalty.
- Remote work can relax time and location constraints, potentially increasing schedule flexibility Chung and Horst (2018).
  - However, this project does not directly observe actual telework use. Instead, it uses occupational teleworkability as a proxy for potential exposure to remote-work flexibility.
- In the Korean context, 정예은 and 홍지훈 (2023) show that workers in teleworkable occupations experienced relatively higher working hours after COVID-19.

Introduction

- COVID-19 generated large and uneven labor-market shocks across workers and occupations.
- The shock was particularly relevant for mothers: Baek and Park (2022) show that women with young children experienced larger reductions in employment and working hours during the first wave of COVID-19.
- At the same time, remote work became more relevant for jobs that could be performed outside the workplace.
- 정예은 and 홍지훈 (2023) show that Korean workers in teleworkable occupations experienced relatively higher working hours after COVID-19.
- This project examines whether mothers in teleworkable occupations experienced relative improvements in labor-market outcomes after COVID-19.

Research Question

Main Question

Did Mothers in Teleworkable Occupations Experience Relative Labor-Market Improvements after COVID-19?

Main Treatment and Comparison Groups

- Group 1: Mothers, defined as women with children
  - Following Goldin (2014), I interpret motherhood not only as a current childcare burden, but also as a cumulative labor-market history involving career interruptions and reduced hours.
- Group 2: Childless women
- Group 3: Men

Empirical Approach

- I compare post-COVID changes across these groups within teleworkable occupations.
- I also conduct subgroup analyses to examine the robustness of the results.

Data Sources

한국노동패널조사 (Korean Labor and Income Panel Study, KLIPS)

- I construct the raw panel from the 19th–27th waves of KLIPS.
- The displayed analysis uses labor-market outcome years from 2016 to 2022.
- Since income questions refer to the previous year, the 26th wave is needed to recover income information for 2022; observations after 2022 are dropped before analysis.
- As KLIPS does not provide consistent information on actual telework use, teleworkability is used as an occupation-level proxy for potential remote-work feasibility.

Teleworkability

- I classify occupations by teleworkability following 최성웅 (2020).
- The classification is based on broad occupational categories.
- Occupations in categories 1–3 are classified as teleworkable, while those in categories 4–9 are classified as less teleworkable.

Table: Classification of Teleworkable Occupations

원격근무 가능	원격근무 어려움
관리자	서비스 종사자
전문가 및 관련 종사자	판매 종사자
사무 종사자	농림어업 숙련 종사자
	기능원 및 관련 기능 종사자
	장치 · 기계조작 및 조립 종사자
	단순노무 종사자

Source: 최성웅 (2020)

Summary Statistics: Main Analysis Sample

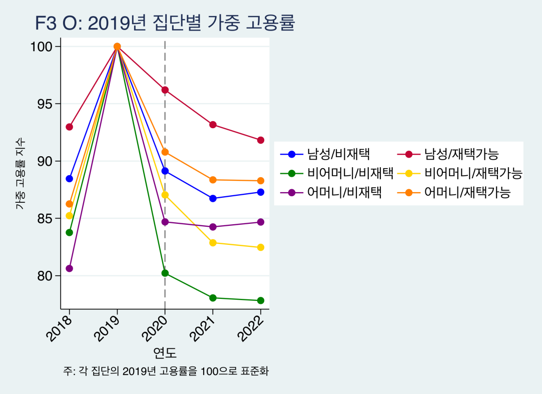
	Mean	SD	Min	Max	N
Panel A: Key Outcomes					
Weekly regular hours	40.41	5.03	6.00	84.00	2,295
Log hourly wage	0.63	0.49	-2.70	2.36	2,293
Log weekly labor income	4.31	0.52	0.99	6.05	2,406
Annual labor income	4,388.50	2,246.72	140.00	22,000.00	2,406
Panel B: Demographics					
Female	0.42	0.49	0.00	1.00	2,409
Age	40.54	9.81	19.00	79.00	2,409
Married	0.69	0.46	0.00	1.00	2,409
Has child	0.63	0.48	0.00	1.00	2,409
College or more	0.86	0.35	0.00	1.00	2,409
Panel C: Spouse Income					
Spouse labor income	4,069.29	3,041.23	80.00	55,000.00	1,165

Weighted: Yes. Mean and SD use 2019 individual weights. Annual and spouse income are measured in 10,000 KRW.

DiD Analysis Sample

- Outcome years used in the DiD analysis: 2016–2022.
  - Average DiD: 2016–2019 vs. 2021–2022.
  - Event study: 2016–2022, with 2019 as the omitted base year.
- Treatment and comparison groups are fixed using 2019 characteristics.
  - Sex, child status, marital status, and occupational teleworkability are measured in 2019.
  - Occupation, industry, and region controls are also fixed at their 2019 values.
- No additional age restriction is imposed.
- For all displayed DiD and event-study results, I restrict the sample to individuals observed and employed in every year from 2020 to 2022.
  - Employed means positive reported weekly regular working hours.
- Decomposition, sub-sample, and placebo exercises use the relevant 2019-defined comparison groups shown in each slide, but keep the same 2020–2022 employment-survivor restriction.
- Log outcome regressions require positive working hours or positive labor income. Spouse-income specifications drop observations with missing spouse income.

Description



- Job retention after COVID-19 differs substantially across groups.
- Post-COVID comparisons may therefore be affected by differential job loss or occupational exit.
- Displayed DiD specifications restrict the sample to workers employed in every year from 2020 to 2022.
- Treatment status and control cells use 2019 occupation, industry, and region.
- The code does not require workers to remain in the same occupation after 2019, so estimates are conditional on employment survival.

Identification Strategy

Difference-in-Differences

$$Y_{it} = \alpha_i + \lambda_t + \beta\left(\text{Treatment}_i \times \text{Post}_t\right) + \gamma_{o(i),t} + \delta_{j(i),t} + \rho_{r(i),t} + \varepsilon_{it}$$

Event Study

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\tau \neq 2019} \beta_\tau \left(\text{Treatment}_i \times \mathbf{1}[t = \tau]\right) + \gamma_{o(i),t} + \delta_{j(i),t} + \rho_{r(i),t} + \varepsilon_{it}$$

- Pre-period: 2016–2019; Post-period: 2021–2022; 2020 is excluded from the DiD.
- Y_{it} is one of three outcomes: log hourly wage, log weekly regular hours, or log weekly labor income.
- The treatment indicator changes by comparison slide, but is always fixed using 2019 group membership.
- Standard errors are clustered by individual.

Outcome Definitions

- KLIPS annual labor income is reported for the previous calendar year.
- I shift income to the actual income year before constructing outcomes.
  - Example: income reported in the 2020 survey is matched to the 2019 observation.
- Weekly labor income is constructed as:

$$\text{weekly labor income}_{it} = \frac{\text{annual labor income}_{it}}{52}$$

- Hourly wage is constructed as:

$$\text{hourly wage}_{it} = \frac{\text{weekly labor income}_{it}}{\text{weekly regular hours}_{it}}$$

- Regression outcomes are:

$$\log(\text{hourly wage}_{it}), \quad \log(\text{weekly regular hours}_{it}), \quad \log(\text{weekly labor income}_{it})$$

Control Variables

- Individual fixed effects: α_i.
- Year fixed effects: λ_t.
- 2019 occupation $\times$ year fixed effects: $\gamma_{o(i),t}$.
- 2019 industry $\times$ year fixed effects: $\delta_{j(i),t}$.
- 2019 region $\times$ year fixed effects: $\rho_{r(i),t}$.
- Weighted specifications use 2019 individual weights.
- Spouse-income specifications control for current-year spouse income, or 2019 spouse income interacted with time.

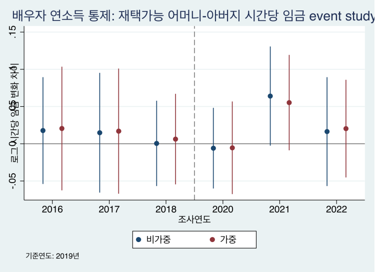
Main Result 1: Mothers vs. Fathers with Spouse Income Control

	Log Hourly Wage		Log Weekly Hours		Log Weekly Income	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Mother × Post	0.0368	0.0287	0.0227*	0.0276**	0.0598**	0.0658**
	(0.0282)	(0.0265)	(0.0120)	(0.0124)	(0.0303)	(0.0272)
Spouse Annual Income	0.000843	-0.000810	-0.000196	-0.000666	0.00153	-0.000605
	(0.00313)	(0.00333)	(0.000811)	(0.000855)	(0.00268)	(0.00266)
Observations	4,377	4,377	4,386	4,386	4,603	4,603
R^2	0.840	0.853	0.741	0.738	0.838	0.854

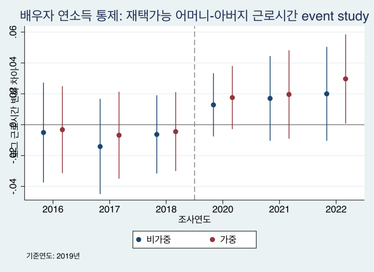
Spouse annual income is measured in KRW 10 million. Observations with missing spouse income are excluded.
Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- The result suggests that mothers' weekly hours increased relative to fathers after COVID-19 among teleworkable workers who remained employed after COVID-19.
- This pattern is consistent with the possibility that teleworkability eased some work-family constraints, but the mechanism is not directly observed.
- Current-year spouse income may raise a bad-control concern. Therefore, I run robustness checks using no spouse-income control and 2019 fixed spouse income.

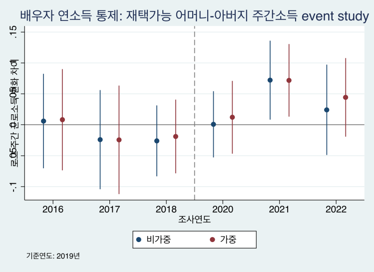
Main Result 1: Event Study with Spouse Income Control



Log Hourly Wage



Log Weekly Hours



Log Weekly Income

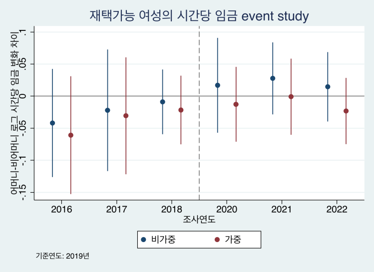
Main Result 2: Teleworkable Mothers vs. Teleworkable Non-mothers

	Log Hourly Wage		Log Weekly Hours		Log Weekly Income	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Mother × Post	0.0298	0.00576	0.0146	0.0244*	0.0404*	0.0344
	(0.0225)	(0.0234)	(0.0105)	(0.0136)	(0.0232)	(0.0240)
Observations	4,594	4,594	4,665	4,665	4,881	4,881
R^2	0.775	0.801	0.777	0.786	0.774	0.793

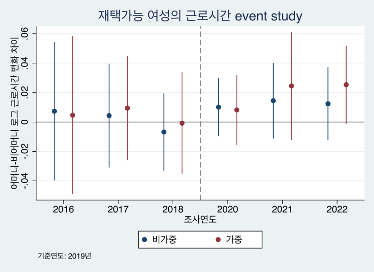
Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- In the broader comparison between teleworkable mothers and teleworkable non-mothers, the estimated coefficients are mostly positive alongside with the theoretical prediction, but statistically weak.
- Therefore, this result provides only limited evidence of a relative improvement for mothers.

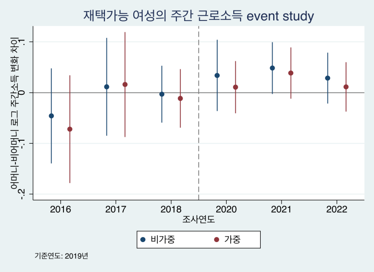
Main Result 2: Event Study



Log Hourly Wage



Log Weekly Hours



Log Weekly Income

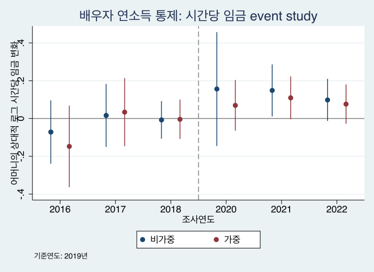
Result 2 Decomposition 1: Married Mothers vs. Married Childless Women

	Log Hourly Wage		Log Weekly Hours		Log Weekly Income	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Mother × Post	0.115**	0.0927**	0.0124	0.0156	0.121**	0.104**
	(0.0465)	(0.0452)	(0.0182)	(0.0233)	(0.0491)	(0.0513)
Spouse Annual Income	0.00152	0.000755	-0.00171	-0.00117	-0.000143	-0.000390
	(0.00636)	(0.00639)	(0.00168)	(0.00147)	(0.00466)	(0.00457)
Observations	2,540	2,540	2,551	2,551	2,722	2,722
R^2	0.788	0.810	0.792	0.788	0.788	0.811

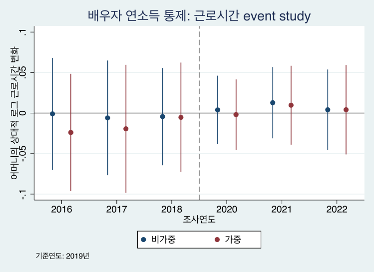
Spouse annual income is measured in KRW 10 million. Observations with missing spouse income are excluded.
Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- Compared with married childless women, married mothers show a sizable relative increase in hourly wage and weekly income, while weekly hours remain stable.
- This suggests that the positive income effect in this comparison is mainly driven by hourly wage rather than labor supply.

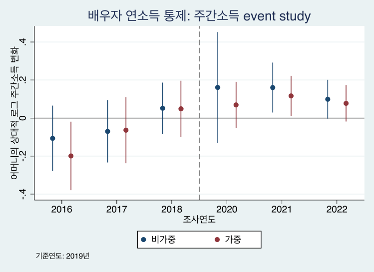
Result 2 Decomposition 1: Event Study with Spouse Income Control



Log Hourly Wage



Log Weekly Hours



Log Weekly Income

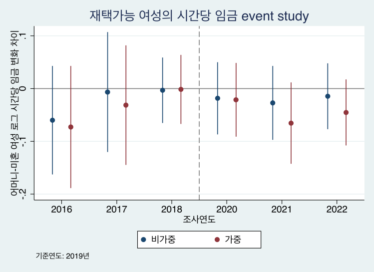
Result 2 Decomposition 2: Married Mothers vs. Unmarried Childless Women

	Log Hourly Wage		Log Weekly Hours		Log Weekly Income	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Mother × Post	-0.0104	-0.0416	0.0122	0.0258	-0.00363	-0.0176
	(0.0282)	(0.0304)	(0.0136)	(0.0186)	(0.0285)	(0.0288)
Observations	3,552	3,552	3,615	3,615	3,777	3,777
R^2	0.790	0.814	0.770	0.794	0.782	0.807

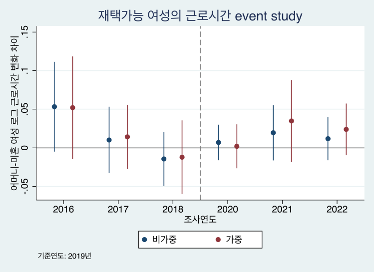
Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- This comparison shows no clear post-COVID relative change for married mothers against unmarried childless women.
- Together with the previous result, this suggests that the positive mother–non-mother gap is partly driven by the relative decline of married childless women.

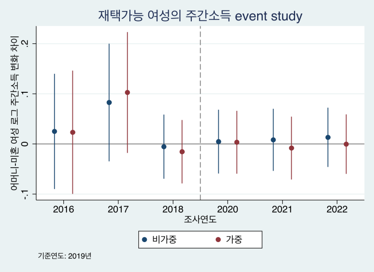
Result 2 Decomposition 2: Event Study



Log Hourly Wage



Log Weekly Hours



Log Weekly Income

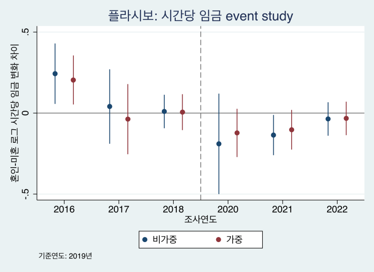
Sub: Married vs. Unmarried Teleworkable Childless Women

	Log Hourly Wage		Log Weekly Hours		Log Weekly Income	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Married × Post	-0.101**	-0.0909*	-0.00954	-0.0162	-0.0639	-0.0511
	(0.0450)	(0.0493)	(0.0126)	(0.0178)	(0.0527)	(0.0576)
Observations	1,211	1,211	1,244	1,244	1,294	1,294
R^2	0.830	0.856	0.892	0.913	0.814	0.815

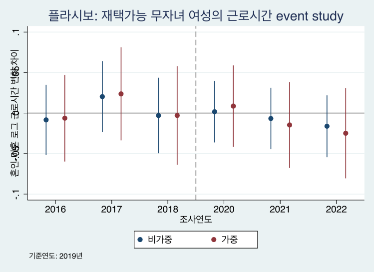
Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- This result indicates that married childless women experienced a relative decline in hourly wage compared with unmarried childless women.

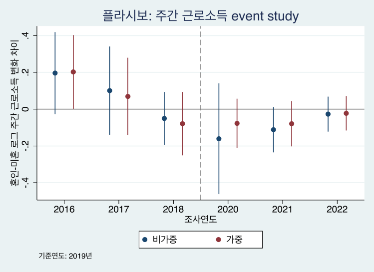
Sub: Event Study



Log Hourly Wage



Log Weekly Hours



Log Weekly Income

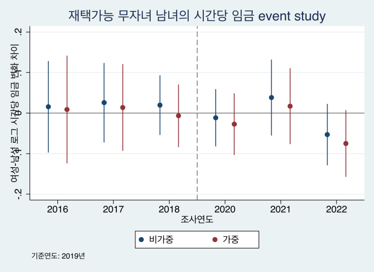
Placebo: Teleworkable Childless Women vs. Childless Men

	Log Hourly Wage		Log Weekly Hours		Log Weekly Income	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Female × Post	-0.0155	-0.0267	0.00204	-0.00917	-0.00343	-0.0221
	(0.0350)	(0.0319)	(0.0110)	(0.0122)	(0.0360)	(0.0339)
Observations	3,435	3,435	3,491	3,491	3,657	3,657
R^2	0.808	0.826	0.805	0.817	0.814	0.821

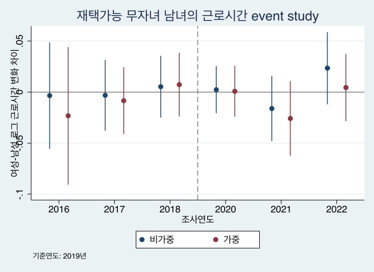
Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- The near-zero estimates suggest that the main patterns are unlikely to be driven by a general female-versus-male COVID shock among childless workers.

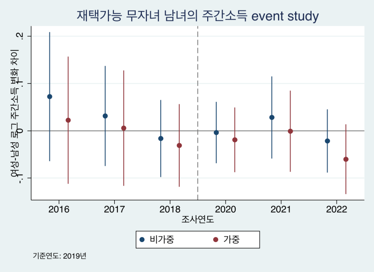
Placebo: Event Study



Log Hourly Wage



Log Weekly Hours



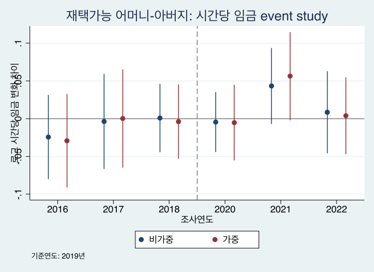
Log Weekly Income

Robustness 1-1: Mothers vs. Fathers without Spouse Income Control

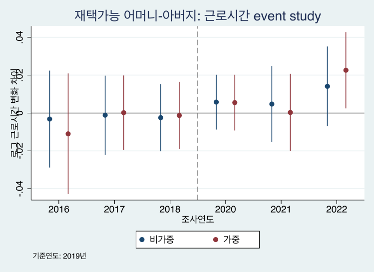
	Log Hourly Wage		Log Weekly Hours		Log Weekly Income	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Mother × Post	0.0305	0.0340	0.0111	0.0135*	0.0496**	0.0617**
	(0.0214)	(0.0227)	(0.00830)	(0.00800)	(0.0229)	(0.0255)
Observations	7,409	7,409	7,498	7,498	7,774	7,774
R^2	0.842	0.851	0.705	0.698	0.836	0.845

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

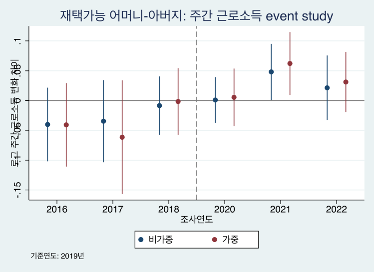
Robustness 1-1: Event Study



Log Hourly Wage



Log Weekly Hours



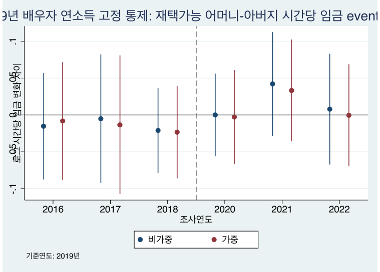
Log Weekly Income

Robustness 1-2: Mothers vs. Fathers with 2019 Spouse Income Control

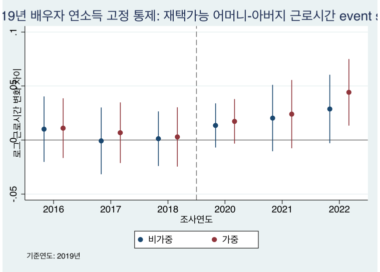
	Log Hourly Wage		Log Weekly Hours		Log Weekly Income	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Mother × Post	0.0341	0.0247	0.0227*	0.0307**	0.0539*	0.0589**
	(0.0288)	(0.0278)	(0.0129)	(0.0124)	(0.0308)	(0.0289)
2019 Spouse Income × Post	0.00120	0.00229	-0.00107	-0.00129	0.00123	0.00183
	(0.00384)	(0.00436)	(0.00145)	(0.00160)	(0.00450)	(0.00552)
Observations	4,390	4,390	4,442	4,442	4,603	4,603
R^2	0.839	0.853	0.733	0.729	0.837	0.854

Spouse income is fixed at its 2019 value and interacted with the post-period indicator.
Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

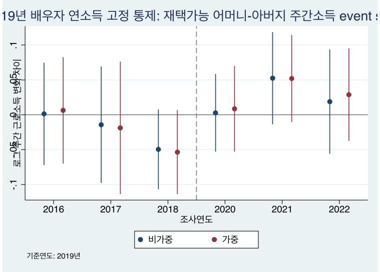
Robustness 1-2: Event Study with 2019 Spouse Income Control



Log Hourly Wage



Log Weekly Hours

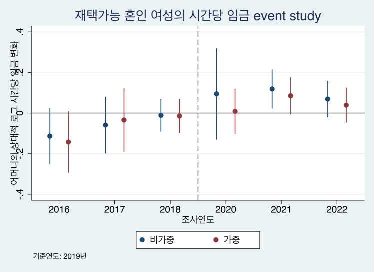


Log Weekly Income

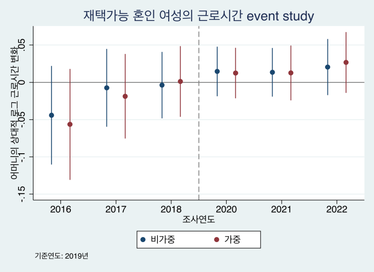
Robustness 2-1: Decomposition 1 without Spouse Income Control

	Log Hourly Wage		Log Weekly Hours		Log Weekly Income	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Mother × Post	0.102***	0.0822**	0.0259*	0.0293	0.115***	0.0960**
	(0.0367)	(0.0354)	(0.0146)	(0.0180)	(0.0384)	(0.0388)
Observations	2,987	2,987	3,033	3,033	3,174	3,174
R^2	0.793	0.815	0.784	0.778	0.792	0.816

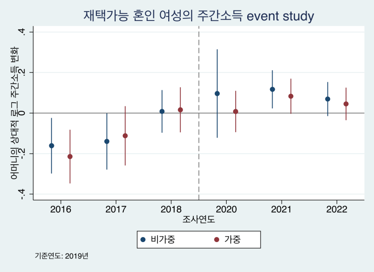
Standard errors in parentheses. $^{*} p < 0.10$, $^{**} p < 0.05$, $^{***} p < 0.01$.



Log Hourly Wage



Log Weekly Hours



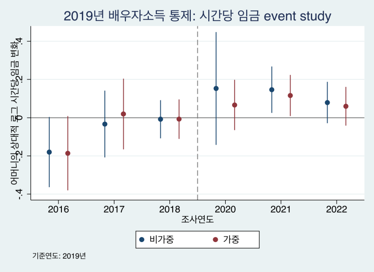
Log Weekly Income

Robustness 2-2: Decomposition 1 with 2019 Spouse Income Control

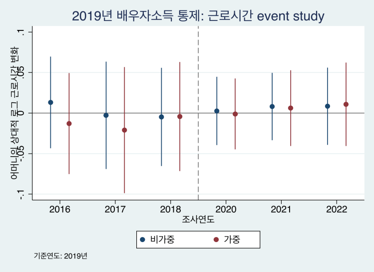
	Log Hourly Wage		Log Weekly Hours		Log Weekly Income	
	Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Mother × Post	0.115**	0.0983**	0.00963	0.0152	0.114**	0.104**
	(0.0460)	(0.0452)	(0.0173)	(0.0217)	(0.0484)	(0.0509)
2019 Spouse Income × Post	0.00183	0.00287	-0.00357**	-0.00265	-0.000241	0.00111
	(0.00476)	(0.00502)	(0.00181)	(0.00180)	(0.00608)	(0.00646)
Observations	2,511	2,511	2,548	2,548	2,680	2,680
R^2	0.784	0.803	0.785	0.780	0.786	0.805

2019 spouse income is measured in KRW 10 million and interacted with the post-period indicator.
Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

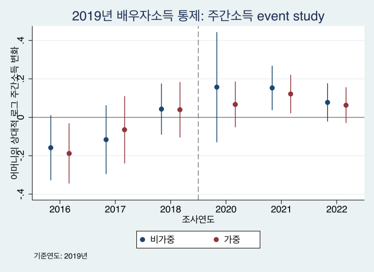
Robustness 2-2: Event Study with 2019 Spouse Income Control



Log Hourly Wage



Log Weekly Hours



Log Weekly Income

Conclusion

- Among teleworkable workers who remained employed after COVID-19, mothers experienced relative post-COVID gains in labor-market outcomes.
- These gains are clearest for weekly income, while evidence for hourly wage and working hours is more mixed.
- Decomposition results suggest that the pattern is partly driven by the relative decline of married childless women.

Limitations & Future Works

Limitations

- Teleworkability is measured at the occupation level and may reflect endogenous occupational selection.
- Restricting the displayed DiD sample to workers employed in 2020–2022 improves comparability, but results are conditional on post-COVID employment survival.
- The analysis does not directly test wage convexity, actual telework use, childcare time, or child-age heterogeneity.
- Teleworkability is potentially correlated with high-skilledness.

Future Work

- Examine wage convexity and long-hour wage premiums more directly.
- Compare mothers by child age.
- Use a more detailed occupational classification (e.g., 3-digit occupation codes) to compare occupations.

References I

- Baek, Jisoo and Wooyoung Park (2022). "COVID-19, Childcare and Women's Labor Supply". In: *The Korean Economic Review* 38.2, pp. 323–345.
- Chung, Heejung and Mariska van der Horst (2018). "Women's Employment Patterns after Childbirth and the Perceived Access to and Use of Flexitime and Teleworking". In: *Human Relations* 71.1, pp. 47–72.
- Goldin, Claudia (2014). "A Grand Gender Convergence: Its Last Chapter". In: *American Economic Review* 104.4, pp. 1091–1119.
- 정예은 and 홍지훈 (2023). "직무별 재택근무 가능성에 따른 코로나19의 고용효과 분석". In: *시장경제연구* 52.2, pp. 81–112.
- 최성웅 (2020). "재택근무가 가능한 일자리의 특성과 분포: 물리적 근로환경을 중심으로". In: *한국경제지리학회지* 23.3, pp. 276–291.